

Price Elasticity of Demand

▼ What is price elasticity of demand?

Price elasticity of demand is the responsiveness of quantity demanded to changes in price, E_d

i.e. The proportionate change in quantity demanded if there is a change in price

▼ How is it calculated?

Calculated as $\text{Elasticity of demand} = \frac{\text{percentage change of quantity demanded}}{\text{Percentage change in price}}$

Straight percentage formula:

Example: Lets say that the price of coffee is \$4 and we want to know how much a 10% change in price is going to affect coffee consumption. Research indicates this change in price would lead to a 30 000 coffee consumption per week decrease from 1 000 000. We want to know how large this fall in quantity is relative to the change in price.

We can use two formulas

1. $E_d = \frac{\text{percentage change of quantity demanded}}{\text{Percentage change in price}}$ and

2. $E_d = \frac{\text{change in } Q}{\text{quantity}} \times \frac{\text{price}}{\text{change in price}}$

1. $\text{Change in } Q = \frac{30\,000}{Q=1\,000\,000} \times P = 4 / \text{Change in } P = 0.40$

$= \frac{30\,000}{1\,000\,000} \times \frac{4}{0.40}$

$= 0.3$

2. $\text{Change in } Q = 3\% / \text{Change in } P = 10\%$

$= 3\% / 10\%$

=0.3

Midpoint formula: $E_d = \frac{\text{Change in } Q}{Q_{\text{ave}}} \times \frac{P_{\text{ave}}}{\text{Change in } P}$

Example: Lets say a good decreases in price from \$11 to \$10, resulting in a change in quantity demanded from 400 000 to 500 000. This means the average price is \$10.50 and (P_{ave}) and the average quantity would be 450 000 (Q_{ave}).

Therefore: $E_d = 100\,000 / 450\,000 \times 10.5 / 1$
= 2.33

Usually economists use the midpoint formula as it can speculate changes across the whole curve

▼ When is PED elastic and inelastic?

If the price elasticity of demand is >1 then it is elastic

If the price elasticity of demand is <1 it is inelastic

Ignore the $-$ symbols

Elastic goods typically have relatively close substitutes such as fast food and fruits

Percentage change in quantity demanded $>$ Percentage change in price

Inelastic goods have relatively little substitutes such as petrol and cigarettes

Percentage change in quantity demanded $<$ Percentage change in price

▼ What about perfectly elastic/inelastic/unitary?

Perfectly elasticity is where $PED = \infty$

If price changes, quantity demanded changes infinitely (to ∞)

This is when there is a perfect market competition
e.g. agricultural market

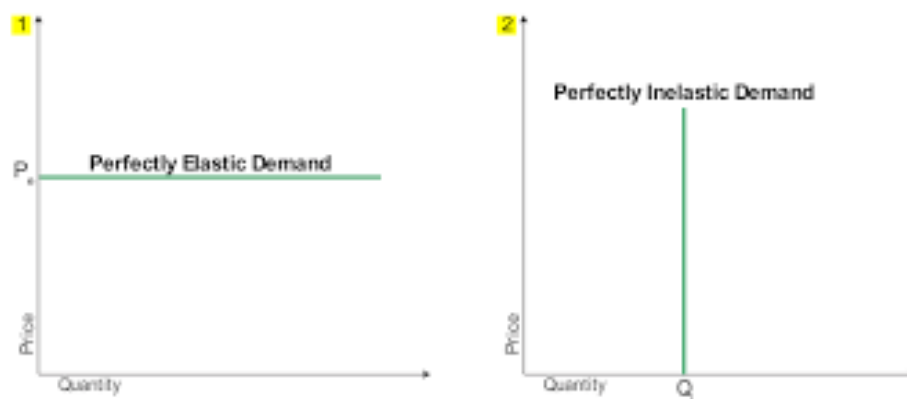
Perfectly inelastic is where $PED = 0$

If price changes then there is no change in quantity demanded

Reflects market where there is fixed demand regardless of price e.g. some medicines

Unitary elasticity is where $PED = 1$

If price increases by 50% then quantity demanded increases by 50%



▼ What are the determinants of PED?

- Share of Budget, income, spent
 - If a large portion of the income is spent then there is a higher tendency to substitute away
 - Cheaper goods: elastic
 - Expensive goods: inelastic
- Number of close substitutes
 - More close substitutes = elastic
 - Less close substitutes = inelastic
- Time

- In the short term it is harder to adjust usually so the demand becomes inelastic
- If you have a longer time its easier to adjust so the demand becomes more elastic
- E.g. increased petrol price in the short term wont effect peoples buying petrol, however, if there is increased price over a long time people will shift towards public transport or hybrid cars
- Scope of the market
 - If its niche or broad
 - Broad = elastic
 - Niche = inelastic

Luxuries are elastic

Necessities are inelastic

▼ What is the impact of PED on total revenue?

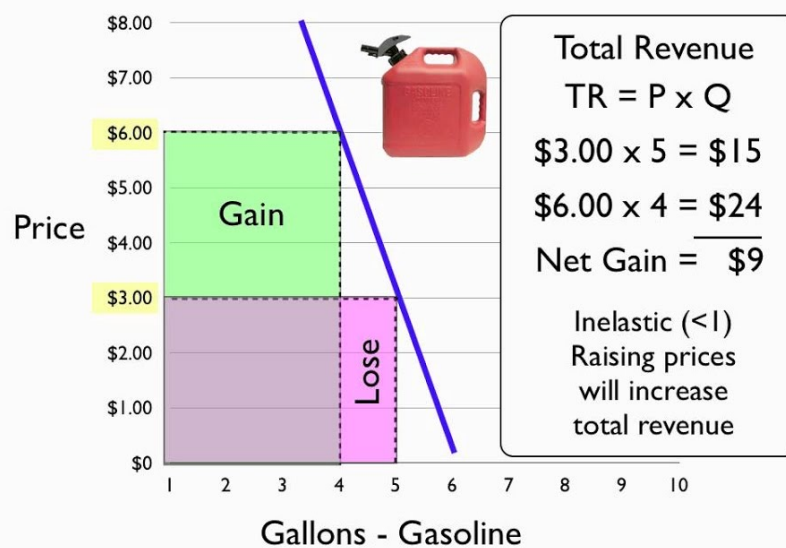
Total revenue = price x quantity

Effects of the change in price on total revenue is dependant on the price elasticity of the product

Price elastic products: total revenue increases as price drops

Price inelastic products: total revenue decreases as price drops

Unitary elastic products: no change in total revenue



▼ What is the significance of PED on tax revenue?

Tax incidence/burden of tax is the person or group who bears the majority of the tax

- If the good is price inelastic in demand then the amount of tax revenue is greater and the burden falls mostly on consumers
 - Relatively small change in quantity demanded
- If the good is price elastic in demand then the amount of tax revenue is less and the burden falls mostly on the supplier.
 - Relatively larger change in quantity demanded

The incidence of a tax on cigarettes – an example

If the government puts an extra tax of £1.00 on each packet of cigarettes, the legal incidence is on the cigarette smoker. However, the local market may be very competitive, with many sellers, so that a retailer may fear they will suffer from lost sales, and decide to put up the price by only 50p, and pay the balance of 50p to the government themselves. In this case, the economic incidence is shared because both are worse off. The smoker is worse off because of the price increase of 50p, and the seller is worse off because 50p must come out of their revenue to pay the government. [Of course, they are not really 'worse off' in the long run if the tax helps them to stop smoking.]

